

Module 2 Reflection Journal

Planning for Access, Agency, and Justice in Invisible Invaders

EDU 486 — Module 2

From Activities to an Access-to-Agency Pedagogy

A reading-to-practice reflection on how coherence, modeling, place, identity, access, technology, and justice shape Piter and Aastha's Invisible Invaders camp.

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Camp Team Invisible Invaders — Piter and Aastha
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Interpretive Lenses JuST Framework; Puzzle Plan and EQUITAS; UDL; PICRAT; justice-centered science
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Inclusion and rigor are the same design problem: both require us to make thinking visible, gather better evidence, and redesign barriers before explaining the learner as the problem.

This journal treats camp pedagogy as an interconnected learning system rather than a list of activities. The **Puzzle Plan** asks me to gather multiple forms of evidence, examine relationships among learners, tasks, tools, and environments, and revise instruction as new evidence appears. The **EQUITAS Access-to-Agency lens** asks whether those conditions create equitable, culturally responsive, identity-safe, and accountable routes for neurodivergent, chronically ill, disabled, multilingual, culturally and linguistically diverse, and multiply marginalized youth to participate, demonstrate competence, and exercise agency.

Journal Focus

At the beginning of Module 2, I had a long list of things I wanted youth to do: collect sand, compare samples, inspect suspected particles, build a model, discuss injustice, and create art for advocacy. I was excited, but I could not explain why those activities belonged together. Reviewing my Heather Table and the readings made me realize that an engaging schedule is not the same as a meaningful learning experience. I want youth to know why each investigation matters, see their knowledge as science, revise ideas with evidence, and decide how to act.

My central principle is that inclusion and rigor are the same design problem. This is personal. As a Latino (Dominican), bilingual, neurodivergent, and chronically ill learner and educator, I know what it is to be praised at my peak and judged when my brain or body cannot perform on demand. I do not want youth to hide struggle or overperform to seem capable. My Puzzle Plan, through the EQUITAS lens, asks me to gather richer evidence and question the environment before the learner; Module 2 turned that commitment into camp decisions.

For Invisible Invaders, I keep returning to the anchoring phenomenon, *“Humans aren’t trash cans, but we have plastics in our bodies,”* and the driving question, *“How does plastic break into tiny pieces, move through the environment and into living bodies, and how can we stop it fairly?”* I want youth to investigate that question genuinely, not chase an answer Aastha and I already chose. Our facilitation will determine whether they help build the science or simply receive our conclusions.

Principles at a Glance

Principle	Reading roots	Visible camp decision
Coherence belongs to learners	Reiser et al.; Cherbow et al.; Wiggins and McTighe	Keep the driving question, complete sequence, and Four Gotta-Haves visible; give one direction at a time; preserve pause and re-entry.
Models are public and revisable	Windschitl and Thompson; Ippolito et al.; Smetana et al.; Wisbrock et al.	Move from controlled jars and field evidence to a source-to-body model; label claims supported, possible, or unknown; revise instead of replacing youth thinking.
Place and community knowledge are science	Seiki et al.; Starr et al.; Nesmith et al.	Treat Charlotte Beach and youth-known places as evidence; examine infrastructure and power; invite community connections without requiring disclosure.
Identity and joy belong inside rigor	Mercier and Carlone; Muhammad	Recognize investigating, designing, connecting, caring, revising, and advocating; notice quiet contributions; provide movement and low-energy routes.
Access must be designed into thinking	Liljedahl; Kimmons et al.; CAST	Preserve demanding sensemaking without requiring one posture, pace, group, or communication mode; use technology only for access, evidence, or agency.
Justice links mechanism, power, and action	Smetana et al.; Wisbrock et al.; Nesmith et al.; Muhammad	Connect plastic pathways to production, infrastructure, policy, and unequal burdens; support collective action without blame or unsupported health claims.

Principle 1: Coherence Must Be Visible to Learners

Reiser, Novak, McGill, and Penuel (2021) argue that a storyline should be coherent from the students' perspective: their questions and explanations should create a real need for the next investigation. Cherbow, McNeill, Lowell, and Grymonpre (2026) helped me see that even strong curriculum cannot create that experience alone. I must help youth connect their questions, our resources, and the explanation we are building. Backward design helps me identify the final understanding and acceptable evidence (Wiggins & McTighe, 2005),¹ but the plan remains mine unless youth can see and influence it.

For Invisible Invaders, coherence must be something youth can see, not something hidden in our planning document. Aastha and I will keep the driving question, day's purpose, activity sequence, and Four Gotta-Haves visible: Earth systems are interconnected; water transports materials; living organisms respond to environmental conditions; and human infrastructure influences natural systems. Each connecting question must help youth explain one of those ideas so they know why the activity belongs.

Meeting the Freedom Scholars and staff on July 21 made this concrete. Youth asked for short explanations, one direction at a time, and a chance to begin quickly. I heard a request for a clearer route, not easier science. A visual agenda, completed example, and brief demonstrations reduce hidden working-memory demands. They also let someone join late, take a break, or return after pain, sensory overload, or attention loss without explaining themselves publicly.

Principle 2: Models Are Public, Revisable Thinking

Before this module, I pictured a model as a polished final product. The modeling toolkit helped me see it as public thinking that can be questioned and changed. Windschitl and Thompson (2013) show how models make visible events and invisible mechanisms available for critique. Science and literacy also strengthen each other when youth investigate and communicate for a real purpose (Ippolito, Condie, Blanchette, & Cervoni, 2018), while local evidence can lead to explanation and public communication (Smetana, Wisbrock, Zaker, & Hoellein, 2019; Wisbrock et al., 2020).

I want our camp model to preserve change, uncertainty, and evidence limits. On Field Friday, youth will compare four controlled jars and Charlotte Beach sand. On Monday, they will build a first source-to-body model and add evidence during the week about transport, friction and wear, living systems, infrastructure, and unequal decisions. On Thursday, they will revise their shared model instead of replacing it with an adult answer. Friday's KEEP, QUESTION, CHANGE routine will support one final evidence-based decision.

I also have to be careful with our claims. A particle that looks like plastic is not chemically confirmed plastic, and finding microplastics in a body or food shows exposure without proving a route or health effect. Youth can label claims as *supported*, *possible*, or *still unknown*. I want them to see uncertainty as honest science, not failure, and to advocate without exaggerating the evidence.

I do not want fast speech or comfort with public presentation to become our evidence of understanding. A learner may speak, point, draw arrows, write a label, dictate to a partner, take an evidence photograph, or pass and return. These are not easier versions of modeling. Every route still connects evidence, mechanism, and claim; only the way thinking becomes visible changes.

Principle 3: Place, Community, and Family Knowledge Are Science

Seiki, Domínguez, and Asato (2021) helped me place family knowledge beside formal science instead of asking youth to leave one world at the classroom door. Starr et al. (2026) show that outdoor learning is not simply science moved outside; place changes relationships among youth, families, histories, and socioecological systems.

¹The Module 2 schedule labels this reading as 2021; the assigned course PDF is the 2005 expanded second edition cited here.

Nesmith et al. (2021) reminded me that environmental problems also involve policy, economics, ethics, art, history, and community knowledge.

Charlotte Beach and the places youth know are not scenery; they are evidence and part of the question of responsibility. During my first semester in the United States, I shared an interpretation and watched the room's attention move from my idea to my accent. I felt my intelligence had to be translated before it could be heard. That memory shapes my work with Freedom Scholars. We will ask who decides, who benefits, who carries burdens, and whose knowledge is recognized. Youth can connect the model to familiar places, stories, languages, or objects without sharing private addresses or being reduced to a ZIP code.

Connection must remain an invitation, not a demand for disclosure. A question about plastic in human bodies can touch illness, disability, food insecurity, environmental racism, family work, or medical trauma. No learner should have to share a diagnosis, body history, or family experience to prove the topic matters. They can use an invented scenario, public place, provided image, or community example. That protects identity while keeping community knowledge inside the science.

Principle 4: Identity, Recognition, and Joy Belong Inside Rigor

Mercier and Carlone (2021) helped me see how identi-beads and badges can make different forms of STEM participation visible. What matters is not the bead, but whether youth recognize their contribution and the group recognizes it too. Muhammad (2023) brings identity, skills, intellect, criticality, and joy together, reminding me that justice-centered learning cannot end with naming what is wrong.

I want every Freedom Scholar to leave with evidence that their contribution mattered to our science community. Aastha and I will notice observation, revision, connection, advocacy, creativity, care, and questions that redirect the group. I especially want us to notice quiet contributions, not only the fastest speaker or most comfortable presenter. A bead, role, or acknowledgment should help youth name something they value in themselves, not rank them or decide what kind of STEM person they can be.

For me, joy is not a prize after rigorous work. Growing up in the Dominican Republic while navigating poverty and the loss of my mother, school was rarely joyful; that history makes joy part of equitable learning for me. In camp, it can appear through curiosity, movement, humor, advocacy art, relationships, and imagining a more just future. The students' feedback made this practical: pair movement and social connection with a seated or lower-energy path, show an example without requiring imitation, and offer meaningful extensions. Joy is accessible only when youth do not have to perform endless energy.

Principle 5: Access Must Be Built Into the Thinking Environment

Liljedahl (2020) made me examine routines that produce compliance without much thinking. I value rich tasks, visible collaborative work, purposeful hints, and responses that return agency. PICRAT asks what relationship learners have with a tool and whether it replaces, amplifies, or transforms practice (Kimmons, Graham, & West, 2020). UDL reminds me to plan multiple ways to engage, perceive information, and express understanding from the beginning (CAST, 2024).

I cannot treat one posture, communication mode, group arrangement, or technology as proof of thinking. A vertical whiteboard may make one learner's idea public while creating pain, fatigue, motor, sensory, or social barriers for another. Random grouping may widen collaboration or remove a stabilizing relationship. A verbal question may reveal reasoning or measure processing speed. I must preserve visible, collaborative thinking while offering standing, seated, tabletop, visual, verbal, partner, photo, and low-energy routes.

I know what it feels like when an access barrier is read as a character flaw. I do not want quietness, fatigue, uneven energy, or another route to become evidence that a Freedom Scholar is not trying. Aastha and I will

check in proactively, give one direction at a time, keep the sequence visible, make public reading and presenting optional, and let youth pass and re-enter. These supports do not lower my expectations; they help me see what youth understand.

Technology must earn its place in our camp. A phone camera can preserve an observation, zoom in on a suspected fiber, or replace unnecessary copying. A shared model can show revisions, and a timer can support transitions. Digital does not automatically mean useful or just. I will ask whether a tool increases access, strengthens evidence, or expands agency. If paper, a magnifier, or conversation works better, we should use it.

Principle 6: Justice Must Move From Evidence to Collective Action

The litter investigations connect local evidence with communication and action, while the other readings keep this from becoming a narrow cleanup lesson (Muhammad, 2023; Nesmith et al., 2021; Smetana et al., 2019; Wisbrock et al., 2020). If I ask only, “What can an individual camper recycle?” I hide product design, labor, wastewater systems, policy, corporate decisions, and unequal exposure. If I describe harm without a path toward change, I may leave youth with fear or helplessness.

Justice-centered science must connect mechanism, power, and action without blaming people carrying the greatest burden. I do not want campers to believe plastic pollution exists only because individuals recycle incorrectly. Our model will identify where plastic is produced, released, transported, encountered by living systems, and shaped by human choices. Youth can advocate through design, reuse and repair, monitoring, public communication, research, policy, or art that shows the future they want.

Our action must also match our evidence. I do not want to frighten youth, make unsupported health claims, or pretend one cleaned sample solves a system problem. Youth can explain what evidence supports, what remains uncertain, who needs better information, and what change could reduce a pathway or distribute responsibility fairly. That is stronger advocacy because it respects the audience and the young scientist.

My Camp Commitments

These principles leave me with a practical set of commitments for Invisible Invaders:

- 1. Plan backward, then revise forward.** I will keep the final explanation and evidence goals clear while changing the route in response to youth questions, observations, access needs, and feedback.
- 2. Keep the storyline visible.** Every investigation will connect to the driving question and a named Gotta-Have, with the purpose and sequence available for learners who need to pause or re-enter.
- 3. Protect scientific uncertainty.** Models and public products will distinguish observation from inference, suspected from confirmed particles, and exposure evidence from claims about route or harm.
- 4. Design equivalent routes into rigor.** Movement and seated options, speaking and non-speaking options, individual thinking and supported collaboration will all lead to evidence-based explanation and action.
- 5. Track whose thinking becomes public.** I will notice whose questions, model relationships, and proposed actions enter the shared record, especially when a learner contributes quietly or outside the expected form.
- 6. Use technology for access, evidence, or agency.** Novelty is not a learning goal. A tool earns its place when it makes meaningful scientific participation possible or more powerful.
- 7. Debrief like a learning scientist.** At the end of each day I will ask what youth evidence changed our plan, which barrier hid someone’s thinking, and what Aastha and I need to redesign before the next session.

Closing Reflection

I am not leaving Module 2 with one perfect method for an inclusive science camp. I am leaving with a more disciplined way to decide. Coherence asks whether youth can shape the journey; modeling, whether their ideas become public and revisable; place, whose knowledge enters science; identity and joy, who belongs; UDL, PICRAT, and my Puzzle Plan, whether the environment reveals or hides thinking; and justice, whether evidence opens a collective path toward change.

Being a teacher and learning scientist means studying the environment before explaining the learner as the problem. That is the teacher I want to be during Invisible Invaders. I want it to show through one direction at a time, a revisable model, honest evidence labels, optional public presentation, proactive check-ins, meaningful community questions, and more than one way to act. I will know we succeeded if youth recognize their questions, evidence, and decisions in the science we built together.

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